CS2005: Database Systems
Lab # 02: Database Login & Basic SQL

Objective:

1. Table – Relational Model
2. Introduction to SQL (DML, DDL, DCL)
3. SQL sample commands and interactions
4. Metadata/ dictionary
5. Learning and practice

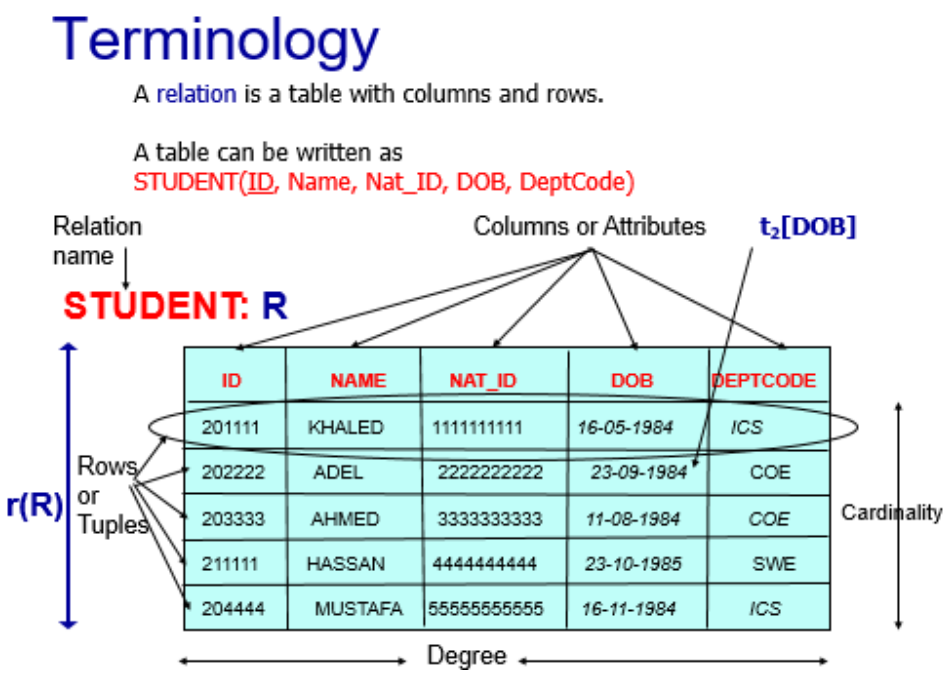
Scope:

The student should know the following:

- Basic Relational Model
- Basic SQL Practices
- Basic exercises

Basic SQL Select Clause with columns
SQL Select with Where Clause using operators =, `IN` and `LIKE`
`ORDER BY` with `ASC` and `DESC` sorting
Use of logical operators: `AND`, `OR`, `NOT` and `IS NULL`

Useful Concepts:



Characteristics of Relation or Table

1. Each relation in the same relational database schema has a distinct name
2. Each value in a tuple is atomic
3. Each attribute in a relation has a distinct name.
4. Values of an attribute are all from the same domain.
5. Each tuple is distinct.
6. Order of attributes has no significance.
7. Order of tuples has no significance, theoretically.

Discussion:

DML, DDL, DCL

```
SELECT TABLE_NAME
FROM INFORMATION_SCHEMA.TABLES;
```

If no rows appear, this means your schema is empty.

To load sample tables with data, download file demobld.sql from gcr.

Open it in SQL Server, execute it and follow up as

```
SELECT   TABLE_NAME, TABLE_TYPE
FROM     EmpDeptDB.INFORMATION_SCHEMA.TABLES
```

If you are already inside database, then just execute

```
SELECT TABLE_NAME
FROM INFORMATION_SCHEMA.TABLES;
```

--To list column names and their datatypes of a table EMP.

```
SELECT
    COLUMN_NAME,
    DATA_TYPE,
    CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'EMP';
```

--To list column names with data of a table EMP

```
Select *
From emp;
```

--From output of above command, choose column names and make SQL as

```
select empno, ename, sal
from emp;

select hiredate
from emp;
```

--To see column names

```
EXEC sp_help emp;
```

--From output of above command, choose column names and make SQL as

```
select dname, loc
from dept;
```

See columns of table emp and their data types



```
SQL> desc emp
 Name                                      Null?    Type
 ----------------------------------------- -------- ------------
 EMPNO                                     NOT NULL NUMBER(4)
 ENAME                                              VARCHAR2(10)
 JOB                                                VARCHAR2(9)
 MGR                                                NUMBER(4)
 HIREDATE                                           DATE
 SAL                                                NUMBER(7,2)
 COMM                                               NUMBER(7,2)
 DEPTNO                                             NUMBER(2)

SQL> select empno, hiredate, deptno, sal
  2  from emp;

     EMPNO HIREDATE      DEPTNO        SAL
---------- --------- ---------- ----------
      7369 17-DEC-80         20        800
      7499 20-FEB-81         30       1600
      7521 22-FEB-81         30       1250
      7566 02-APR-81         20       2975
      7654 28-SEP-81         30       1250
      7698 01-MAY-81         30       2850
      7782 09-JUN-81         10       2450
      7788 19-APR-87         20       3000
      7839 17-NOV-81         10       5000
      7844 08-SEP-81         30       1500
      7876 23-MAY-87         20       1100

     EMPNO HIREDATE      DEPTNO        SAL
---------- --------- ---------- ----------
      7900 03-DEC-81         30        950
      7902 03-DEC-81         20       3000
      7934 23-JAN-82         10       1300

14 rows selected.

SQL>
```



Start writing SQL by looking at above columns

Exercises:

```
exec sp_help dept

select dname, loc
from dept;

select ename, hiredate, job, comm, sal
from emp;

select distinct job
from emp;

select job
from emp;
```

How a question can be asked?

Question: Display job, hire date, salary of employees, order by department number?

Solution:

First to find out an exact table or tables that provides required columns by writing the following SQL

```
SQL>select * from tab;
```

Then to see an exact name of column(s) from table (in this case table you decided is EMP), run command as

```
SQL>exec sp_help dept
```

Based upon above SQL statements formulate SQL statement as follows

```
SQL>select job, hiredate, sal
from emp
order by deptno;
```

Q. List of employee name, hiring date, job title, commission and salary of those employees who are clerks.

```
select ename, sal, deptno
from emp
order by ename;

select job, deptno, sal
from emp
order by job asc, deptno desc;

select ename, hiredate, job, comm, sal
from emp
where job='CLERK';

--string enlcosed in single quotation is --case sensitive

select ename, hiredate, job, comm, sal from emp
where job='CLERK'
and sal>1000;

--FOR WHERE use operator IN, BETWEEN, LIKE, IS NULL
--List of employees who may clerk, manager,
--analyst having salary below 1200.

select *
from emp
where job='CLERK' or
job='MANAGER' or
job='ANALYST'
OR SAL<1200;

select *
from emp
where job IN ('CLERK', 'MANAGER', 'ANALYST')
and sal<1200;
```

```
select *
from emp
where (job='CLERK' and SAL<1200) or
(job='MANAGER' and SAL<1200) or
(job='ANALYST' and SAL<1200)
;

select *
from emp
where (job='CLERK' or
job='MANAGER' or
job='ANALYST')
and SAL<1200;

Q.List of employees having salary ranges from 1000 to 3000
select *
from emp
where sal between 1000 and 3000;

select *
from emp
where sal >=1000 and sal <= 3000;

select * from emp
where deptno in (10,30);
```

LIKE (% and _)

```
select *
from emp
where ename like 'M%'
order by ename;

select *
from emp
where ename like 'MAR%';

select *
from emp
where ename like 'MARTI__';

select *
from emp
where ename like '_A%';

select *
from emp
where ename like '%A%';

select *
from emp
where ename NOT like '%A%';

select *
from emp
where comm IS NULL;

select *
from emp
where comm = NULL;  -- does not give you any output
```

Exercise and Logical Evaluation:

```
SQL> select *
  2  from emp
  3  where job='CLERK'
  4  or Job='MANAGER'
  5  and SAL<1200;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM     DEPTNO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7369 SMITH      CLERK           7902 17-DEC-80        700                    20
      7876 ADAMS      CLERK           7788 12-JAN-83       1100                    20
      7900 JAMES      CLERK           7698 03-DEC-81        950                    30
      7934 MILLER     CLERK           7782 23-JAN-82       1300                    10

SQL>
SQL> select *
  2  from emp
  3  where job='CLERK'
  4  or (Job='MANAGER'
  5  and SAL<1200);

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM     DEPTNO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7369 SMITH      CLERK           7902 17-DEC-80        700                    20
      7876 ADAMS      CLERK           7788 12-JAN-83       1100                    20
      7900 JAMES      CLERK           7698 03-DEC-81        950                    30
      7934 MILLER     CLERK           7782 23-JAN-82       1300                    10

SQL>
SQL> select *
  2  from emp
  3  where (job='CLERK'
  4  or Job='MANAGER')
  5  and SAL<1200;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM     DEPTNO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7369 SMITH      CLERK           7902 17-DEC-80        700                    20
      7876 ADAMS      CLERK           7788 12-JAN-83       1100                    20
      7900 JAMES      CLERK           7698 03-DEC-81        950                    30
```

Display employee name, job title and gross salary of all employees.

Display employee name, job title, joining date and percentage of commission on salary.

Display employee number, name, job, salary and zakat of 2.5% on salary.

Display different job titles of employees.

Display employee number, name, job, salary and gross salary with the raise of 15% salary with commission.

Display 40% house rent of basic salary of employee, medical allowance 12% of basic salary minus commission, transport allowance 5% of basic salary plus commission.

Display gross salary including house, medical, commission and transport allowances.

Display column alias in select clause.

Display row num or serial no with records and rows.

Use Alias of table name to display columns with Alias.

Make few SQLs with operators IN, (=, <>, >=, <=) and LIKE using WHERE Clause.

Using dictionary to find the constraints on tables, who created the table etc.

Display project number, project name and project manager number.

List of project number, employee number and duration of employee who works in project.

Evaluation:

Your Lab Work grade will depend** on your **active participation**, **individual efforts in solving Lab Problem and Seriousness during the lab.